

12 Path integrals

12.1 Path-integral description of quantum mechanics

In previous lectures, we discussed how to propagate a wavefunction using the propagator operator $\hat{U}(t)$, which can be broken up into N steps of length $\epsilon = t/N$.

$$\begin{aligned}\langle x|\Psi(t)\rangle &= \langle x|\hat{U}(t)|\Psi(0)\rangle = \langle x|\hat{U}(\epsilon)\cdots\hat{U}(\epsilon)|\Psi(0)\rangle \\ &= \int \langle x|\hat{U}(t)|x_0\rangle \langle x_0|\Psi(0)\rangle dx_0 \\ &= \int K(x_0, x, t) \langle x_0|\Psi(0)\rangle dx_0\end{aligned}$$

Here we inserted an identity (integral over a complete set of position states) and defined the position-state matrix elements of the propagator $K(x_0, x, t) = \langle x|\hat{U}(t)|x_0\rangle = \langle x|\hat{U}(\epsilon)\cdots\hat{U}(\epsilon)|x_0\rangle$.

Next we inserting complete sets of states between all short-time propagator operators (and rename the final position x to x_N for later convenience):

$$\begin{aligned}K(x_0, x_N, t) &= \langle x_N|\hat{U}(\epsilon)\cdots\hat{U}(\epsilon)|x_0\rangle \\ &= \int \cdots \int \langle x_N|\hat{U}(\epsilon)|x_{N-1}\rangle \langle x_{N-1}|\cdots|x_1\rangle \langle x_1|\hat{U}(\epsilon)|x_0\rangle dx_1 \cdots dx_{N-1}\end{aligned}$$

Therefore we need to find the representation of the short-time propagator in the position basis and can use the Trotter approximation introduced earlier in the course:

$$\begin{aligned}\langle x_{n+1}|\hat{U}(\epsilon)|x_n\rangle &= \langle x_{n+1}|e^{-i\hat{H}\epsilon/\hbar}|x_n\rangle \approx \langle x_{n+1}|e^{-i\hat{V}\epsilon/2\hbar}e^{-i\hat{T}\epsilon/\hbar}e^{-i\hat{V}\epsilon/2\hbar}|x_n\rangle \\ &= e^{-iV(x_{n+1})\epsilon/2\hbar} \langle x_{n+1}|e^{-i\hat{T}\epsilon/\hbar}|x_n\rangle e^{-iV(x_n)\epsilon/2\hbar}\end{aligned}$$

where we have used the fact that $\hat{V}|x\rangle = V(x)|x\rangle$, i.e. the position state is an eigenstate of the potential operator (like all operators of positions). Finally, we can treat the kinetic-energy part using the free particle result:

$$\begin{aligned}\langle x_{n+1}|e^{-i\hat{T}\epsilon/\hbar}|x_n\rangle &= \int \langle x_{n+1}|e^{-i\hat{T}\epsilon/\hbar}|p\rangle \langle p|x_n\rangle dp \\ &= \int \langle x_{n+1}|p\rangle e^{-ip^2\epsilon/2m\hbar} \langle p|x_n\rangle dp \\ &= \frac{1}{2\pi\hbar} \int e^{ipx_{n+1}/\hbar} e^{-ip^2\epsilon/2m\hbar} e^{-ipx_n/\hbar} dp \\ &= \frac{1}{2\pi\hbar} \int e^{-ip^2\epsilon/2m\hbar + ip(x_{n+1}-x_n)/\hbar} dp \\ &= \frac{1}{2\pi\hbar} \sqrt{\frac{2\pi m\hbar}{i\epsilon}} e^{im(x_{n+1}-x_n)^2/2\epsilon\hbar} \\ &= \sqrt{\frac{m}{2\pi i\epsilon\hbar}} e^{im(x_{n+1}-x_n)^2/2\epsilon\hbar}\end{aligned}$$

So, the total propagator is

$$\begin{aligned}
K(x_0, x_N, t) &= \int \cdots \int \langle x_N | \hat{U}(\epsilon) | x_{N-1} \rangle \langle x_{N-1} | \cdots | x_1 \rangle \langle x_1 | \hat{U}(\epsilon) | x_0 \rangle dx_1 \cdots dx_{N-1} \\
&= \int \cdots \int e^{-iV(x_N)\epsilon/2\hbar} \langle x_N | e^{-i\hat{T}\epsilon/\hbar} | x_{N-1} \rangle e^{-iV(x_{N-1})\epsilon/2\hbar} \\
&\quad \times e^{-iV(x_{N-1})\epsilon/2\hbar} \langle x_{N-1} | \cdots | x_1 \rangle e^{-iV(x_1)\epsilon/2\hbar} \\
&\quad \times e^{-iV(x_1)\epsilon/2\hbar} \langle x_1 | e^{-i\hat{T}\epsilon/\hbar} | x_0 \rangle e^{-iV(x_0)\epsilon/2\hbar} dx_1 \cdots dx_{N-1} \\
&= \left(\frac{m}{2\pi i \epsilon \hbar} \right)^{N/2} \int \cdots \int e^{-iV(x_N)\epsilon/2\hbar} e^{im(x_N - x_{N-1})^2/2\epsilon\hbar} e^{-iV(x_{N-1})\epsilon/2\hbar} \\
&\quad \times e^{-iV(x_{N-1})\epsilon/2\hbar} e^{im(x_{N-1} - x_{N-2})^2/2\epsilon\hbar} \cdots e^{im(x_2 - x_1)^2/2\epsilon\hbar} e^{-iV(x_1)\epsilon/2\hbar} \\
&\quad \times e^{-iV(x_1)\epsilon/2\hbar} e^{im(x_1 - x_0)^2/2\epsilon\hbar} e^{-iV(x_0)\epsilon/2\hbar} dx_1 \cdots dx_{N-1} \\
&= \left(\frac{m}{2\pi i \epsilon \hbar} \right)^{N/2} \int \cdots \int e^{iS_N(x_0, x_1, \dots, x_{N-1}, x_N)/\hbar} dx_1 \cdots dx_{N-1}
\end{aligned}$$

where the discretized action is

$$S_N(x_0, x_1, \dots, x_{N-1}, x_N) = \sum_{n=0}^{N-1} \frac{m(x_{n+1} - x_n)^2}{2\epsilon} - \frac{\epsilon}{2} V(x_0) - \sum_{n=1}^{N-1} \epsilon V(x_n) - \frac{\epsilon}{2} V(x_N)$$

In the infinite N limit, we have a path integral which is written

$$= \int_{x(0)=x_0}^{x(t)=x_N} e^{iS/\hbar} \mathcal{D}x$$

where the action along a given path is

$$S = \int_0^t \left[\frac{m\dot{x}(t')^2}{2} - V(x(t')) \right] dt' = \int_0^t \mathcal{L}(x(t'), \dot{x}(t')) dt'.$$

The probability for a particle to move from x_0 to x_N in time t can thus be written as $|K(x_0, x_N, t)|^2$, which is computed by considering all possible paths and summing over the phase associated with them. If two paths have completely different values of the action, the phases might cancel out, so the dominant contribution to the probability will come from paths which have similar values of the action to their neighbours. We are therefore interested in paths for which a small change to the path results in no change to the action. These are called paths of stationary action. To compute them, we use the mathematical technique called “calculus of variations”.

Consider a path $x(t')$ and a neighbouring path $x(t') + \delta x(t')$. Note that in order to obey the boundary conditions (start at x_0 and end at x_N), we require that $\delta x(0) = \delta x(t) = 0$. The actions

of these two paths are

$$\begin{aligned}
 S[x] &= \int_0^t \mathcal{L}(x, \dot{x}) dt' \\
 S[x + \delta x] &= \int_0^t \mathcal{L}(x + \delta x, \dot{x} + \delta \dot{x}) dt' \\
 &= \int_0^t \left[\mathcal{L}(x, \dot{x}) + \frac{\partial \mathcal{L}}{\partial x} \delta x + \frac{\partial \mathcal{L}}{\partial \dot{x}} \delta \dot{x} \right] dt' \\
 &= S[x] + \int_0^t \left[\frac{\partial \mathcal{L}}{\partial x} \delta x + \frac{\partial \mathcal{L}}{\partial \dot{x}} \delta \dot{x} \right] dt'
 \end{aligned}$$

such that the change in action is

$$\begin{aligned}
 \delta S &= S[x + \delta x] - S[x] \\
 &= \int_0^t \left[\frac{\partial \mathcal{L}}{\partial x} \delta x + \frac{\partial \mathcal{L}}{\partial \dot{x}} \delta \dot{x} \right] dt' \\
 &= \left[\frac{\partial \mathcal{L}}{\partial \dot{x}} \delta x \right]_0^t + \int_0^t \left[\frac{\partial \mathcal{L}}{\partial x} \delta x - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) \delta x \right] dt'
 \end{aligned}$$

The change in the action is 0 if

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{\partial \mathcal{L}}{\partial x}$$

which is recognized to be the same as the Euler–Lagrange equation, from which classical mechanics can be derived.

In the limit that $\hbar \rightarrow 0$ (relative to other system parameters) the paths of stationary action dominate the path integral more and more, and thus we have found a simple link between quantum mechanics and classical mechanics. However, for finite values of $\hbar$, the other paths also contribute and can be used to explain quantum phenomena.

12.2 Path-integral quantum statistics

It is difficult to use path integrals to simulate quantum dynamics on a computer because of the enormous problems related to averaging over terms with different signs (or phases). However, the path-integral approach has been very useful in theoretical chemistry as a tool for computing quantum statistical properties. For instance, the partition function

$$Z = \text{Tr} \left[e^{-\beta \hat{H}} \right] = \int \rho(x, x) dx$$

where the density matrix is defined as

$$\rho(x', x'') = \langle x'' | e^{-\beta \hat{H}} | x' \rangle$$

Note the similarity between the density matrix and the propagator. If we choose $t = -i\beta\hbar$, they are identical, and we can thus think of the density matrix as an imaginary-time propagator.

It is not hard to convert all of the path-integral formulas to imaginary time.

$$\begin{aligned}\rho(x_0, x_N) &= K(x_0, x_N, -i\beta\hbar) \\ &= \left(\frac{m}{2\pi\eta\hbar}\right)^{N/2} \int \dots \int e^{-\bar{S}_N(x_0, x_1, \dots, x_{N-1}, x_N)/\hbar} dx_1 \dots dx_{N-1}\end{aligned}$$

using $t \rightarrow -i\beta\hbar$ and therefore $\epsilon \rightarrow -i\eta$, where $\eta = \beta\hbar/N$. The action becomes

$$\begin{aligned}S_N(x_0, x_1, \dots, x_{N-1}, x_N) &= -\sum_{n=0}^{N-1} \frac{m(x_{n+1} - x_n)^2}{2i\eta} + \frac{i\eta}{2}V(x_0) + \sum_{n=1}^{N-1} i\eta V(x_n) + \frac{i\eta}{2}V(x_N) \\ \bar{S}_N(x_0, x_1, \dots, x_{N-1}, x_N) &= -iS_N(x_0, x_1, \dots, x_{N-1}, x_N) \\ &= \sum_{n=0}^{N-1} \frac{m(x_{n+1} - x_n)^2}{2\eta} + \frac{\eta}{2}V(x_0) + \sum_{n=1}^{N-1} \eta V(x_n) + \frac{\eta}{2}V(x_N)\end{aligned}$$

Thus

$$\rho(x', x'') = \int_{x(0)=x'}^{x(\beta\hbar)=x''} e^{-\bar{S}/\hbar} \mathcal{D}x(u) \quad \bar{S} = \int_0^{\beta\hbar} \left[\frac{1}{2}m\dot{x}(u)^2 + V(x(u)) \right] du$$

and the partition function is

$$\begin{aligned}Z &= \int \rho(x_0, x_0) dx_0 \\ &= \int \int_{x(0)=x_0}^{x(\beta\hbar)=x_0} e^{-\bar{S}/\hbar} \mathcal{D}x(u) dx_0 \\ &= \oint e^{-\bar{S}/\hbar} \mathcal{D}x(u)\end{aligned}$$

In a similar way, we can compute averages of operators which depend on position operators:

$$\begin{aligned}\langle \hat{A} \rangle &= \frac{1}{Z} \text{Tr} \left[e^{-\beta\hat{H}} \hat{A} \right] \\ &= \frac{1}{Z} \int \langle x_0 | e^{-\beta\hat{H}} \hat{A} | x_0 \rangle dx_0 \\ &= \frac{1}{Z} \int \langle x_0 | e^{-\beta\hat{H}} | x_0 \rangle A(x_0) dx_0 \\ &= \frac{1}{Z} \int \rho(x_0, x_0) A(x_0) dx_0 \\ &= \frac{1}{Z} \int \int_{x(0)=x_0}^{x(\beta\hbar)=x_0} e^{-\bar{S}/\hbar} \mathcal{D}x(u) A(x_0) dx_0 \\ &= \frac{1}{Z} \oint e^{-\bar{S}/\hbar} A(x_0) \mathcal{D}x(u)\end{aligned}$$

An alternative way of writing the path-integral formulas shows why the method is so useful.

$$\begin{aligned} Z &= \left(\frac{m}{2\pi\eta\hbar} \right)^{N/2} \int \dots \int e^{-S_N(x_N, x_1, \dots, x_N)/\hbar} dx_1 \dots dx_N \\ &= \left(\frac{m}{2\pi\beta_N\hbar^2} \right)^{N/2} \int e^{-\beta_N U_N(\mathbf{x})} d\mathbf{x} \end{aligned}$$

where $\mathbf{x} = \{x_1, \dots, x_N\}$ is a *ring polymer* (so called because each bead is connected to its neighbour in a ring by springs). This expression looks just like a multidimensional classical ensemble average in which the effective inverse temperature is $\beta_N = \beta/N$ and the effective potential is

$$U_N(\mathbf{x}) = \frac{S_N}{\beta_N\hbar} = \sum_{n=1}^N \frac{1}{2} m \frac{1}{\beta_N^2 \hbar^2} (x_n - x_{n-1})^2 + \sum_{n=1}^N V(x_n).$$

We can also choose to write

$$Z = (2\pi\hbar)^{-N} \iint e^{-\beta_N H_N} d\mathbf{x} d\mathbf{p}$$

where the effective Hamiltonian is

$$H_N = \sum_{n=1}^N \frac{p_n^2}{2m} + U_N(\mathbf{x})$$

and we have introduced effective momenta for each bead $\mathbf{p} = \{p_1, \dots, p_N\}$.

It is easy to show that by taking $N = 1$ in all these formulas, one recovers classical statistics (as in the first lecture). In the $N \rightarrow \infty$ limit, the path-integral approach gives the exact quantum statistical results, and for many chemical systems a good approximation is provided by a small but finite value for N . By consideration of the harmonic oscillator, one sees that the result will only be converged if $\beta_N \hbar \omega \ll 1$, such that for typical simulations such as liquid water at room temperature, $N = 32$ is sufficient.

12.3 Path-integral rate theory

Now we have an effective ring-polymer Hamiltonian, we can define ring-polymer dynamics using Hamilton's equations of motion.

$$\dot{x}_n = \frac{p_n}{m} \quad \dot{p}_n = -\frac{m}{\beta_N^2 \hbar^2} (2x_n - x_{n-1} - x_{n+1}) - \nabla V(x_n)$$

Note that the Hamiltonian has been derived by analogy between quantum and classical *statistics*, and we should not assume that these dynamics also describe quantum *dynamics* correctly. However, in some situations, it turns out that it does quite a good job. We can for instance look at the motion of the centre of mass of the ring-polymer, called the “centroid”:

$$\begin{aligned} \bar{x} &= \frac{1}{N} \sum_{n=1}^N x_n & \bar{p} &= \frac{1}{N} \sum_{n=1}^N p_n \\ \dot{\bar{x}} &= \frac{\bar{p}}{m} & \dot{\bar{p}} &= \frac{1}{N} \sum_{n=1}^N \nabla V(x_n) \end{aligned}$$

Note that in the special case of a harmonic oscillator or free particle the force on the centroid is equal to $\nabla V(\bar{x})$ (because the gradient is linear or non-existent) and will thus move as if it were a classical particle. This is actually good news, as we know that classical mechanics is basically correct for these systems. In more complicated problems, $\nabla V(x_n) \neq \nabla V(\bar{x})$, so the dynamics will be different from classical (good because classical mechanics is wrong), but unfortunately not necessarily the same as quantum.

Something that it does do well is calculating rates. We simply use the ring-polymer Hamiltonian in the flux-side correlation function

$$C_{\text{fs}}(t) = (2\pi\hbar)^{-N} \iint \delta[\bar{x} - x^\ddagger] \frac{\bar{p}}{m} \theta(\bar{x}_t - x^\ddagger) e^{-\beta_N H_N} d\mathbf{x} d\mathbf{p}$$

and then can define the ring-polymer molecular dynamics (RPMD) rate:

$$k_{\text{RPMD}} = \lim_{t \rightarrow \infty} \frac{C_{\text{fs}}(t)}{Z_r}$$

and by analogy to the classical case, we define an ring-polymer TST:

$$k_{\text{RPTST}} = \lim_{t \rightarrow 0_+} \frac{C_{\text{fs}}(t)}{Z_r}$$

Note that RPTST can be (in some cases greatly) improved by using a better dividing surface than the one suggested here.

12.4 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Path integrals

- Feynman, QED: the strange case of light and matter [*easy reading which emphasizes concepts rather than maths*]
- Feynman, Path integrals and quantum mechanics [*classic*]

12.5 What are you expected to be able to do in the oral exam?

Explain how to calculate the rate of a chemical reaction from first principles, describing and assessing the approximations used. In particular:

- Compare classical vs quantum mechanics (correlation functions, equilibrium averages, partition functions, etc.);
- Describe Born-Oppenheimer approximation and how to go beyond it;
- Describe how molecules interact with light in terms of wavefunction propagation;
- Explain how the quantum dynamics of a two-level system differs from that of a system with a continuum of states;

- Define Eyring TST and describe how to evaluate it;
- Describe approximations of TST and how to go beyond them (friction, nonadiabatic effects, microcanonical ensembles, tunnelling, etc.).